



Md. Amir Hamza

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CAREER OBJECTIVE

Ambitious CSE graduate with expertise in network management and AI, seeking to apply technical skills and a strong analytical mindset in a dynamic organization. Committed to continuous learning and contributing effectively to technology initiatives.

EDUCATIONAL QUALIFICATION

Bachelor of Science (B.Sc.) in Computer Science and Engineering	2020-2025
- National Institute of Textile Engineering & Research (NITER), Constitute of DU - CGPA: 3.36 (Out of 4.00) [Till 7th Semesters]	
Higher Secondary Certificate (H.S.C)	2017-2019
- GOVT. MOLLARTEK UDAYAN HIGHER SECONDARY SCHOOL - Dhaka Board - GPA: 4.33 (out of 5.00)	
Secondary School Certificate	2015-2017
- Bangamata Sheikh Fazilatunnesa Mujib Govt. Secondary School - Dhaka Board - GPA: 4.86 (out of 5.00)	

Work Experience

Assistant Project Manager

Software Seba, Borobag, Mirpur-2, Dhaka-1216, Bangladesh

25/02/2025 – 30/03/2025

PROJET WORK

- 1. "Deep Learning Based Fabric Defect Detection Using YOLO & Vision Transformer"**
 - Developed a hybrid model combining YOLOv8 for real-time defect localization and Vision Transformers for refined classification, achieving high accuracy in textile defect detection.
- 2. Bone Fracture Detection Using Deep Learning**
 - Designed and trained a CNN-based deep learning model to detect bone fractures in X-ray images with high accuracy. Applied preprocessing, augmentation, and model optimization techniques to improve diagnostic performance.
- 3. Flood Detection with Explainable AI**
 - Designed a deep learning model using environmental and hydrological data to predict flood events. Applied SHAP-based explainability to identify top contributing features, enhancing interpretability for disaster management decisions.
- 4. Online Voting system**
 - Developed a secure Android-based online voting system with voter authentication, real-time vote casting, and result tracking. Integrated Firebase for backend data handling and user verification.
- 5. Smart Fish Detection System Using Arduino**
 - Built a low-cost fish detection device using Arduino UNO and ultrasonic sensors to help fishermen detect fish presence in water. Displayed real-time results on an LCD screen to support efficient fishing.

Research Work

1. **Advanced Hydro-Informatic Modeling through Feedforward Neural Network, Federated Learning, and Explainable AI for Enhancing Flood Prediction**
IEEE Open Journal of the Computer Society · Mar 31, 2025
<https://ieeexplore.ieee.org/abstract/document/10946125>
2. **Hydro-Informatic Modeling for Flood Prediction through Explainable AI to Interpret Water Dynamics in Bangladesh Perspective**
IEEE · May 23, 2024
<https://ieeexplore.ieee.org/document/10534521>

TRAINING & SEMINAR

- Completed 3 days workshop on microcontroller.
- Attend seminar on Amazon Web service
- Web Development course from Programming Hero
- Attend CS50 program

SKILLS

Expertise : Java, Python, C, C++, MySQL, AI, Machine Learning, Deep learning, Routing & Switching
Comfortable : JavaScript, Selenium, CSS, Bootstrap, Tailwind, Network Security
Familiar : Django, Assembly Language, AWS VPC, Azure Virtual Network,
Other : TensorFlow, Git, GitHub, Bash Scripting, Virtualization (VMware), Odoo

INDUSTRIAL VISIT

- Bangladesh Research and Education Network (BdREN) (2023)

LANGUAGE PROFICIENCY

- Fluent Reading, writing and speaking skills in **Bengali**.
- Moderate Reading, writing and speaking skills in **English**.

EXTRA CURRICULAR ACTIVITIES

- Achieved Univ Gold Award for **“WRS - IML: Waterlogging Reduction System based on IoT and Machine Learning - A Bangladeshi Point of View”**.
- Participant of Hult Prize 2020.
- Participant of Nasa Space Apps Challenge in 2020.
- Volunteer at Society for the Popularization of Science, Bangladesh
- Volunteer at Bangladesh Flying Labs
- Volunteer at NEST Business Incubation Hub

PERSONAL INFORMATION

Father's name: Md. Abdul Aziz

Mother's name: Sultana Binta Kabir

Permanent address: Rajabari, Mirzapur, Tangail, Dhaka

Nationality: Bangladeshi

Date of Birth: 28/12/2001

Marital Status: Unmarried

Religion: ISLAM

Contact: +8801925210455, amirhamzacse2020@gmail.com , [LinkedIn](#)

REFERENCE

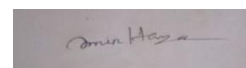
Umme Sara Assistant Professor, Department of Computer Science & Engineering(CSE) National Institute of Textile Engineering & Research(NITER) Mail : usara@niter.edu.bd Mobile : +880-1674 668366	Md. Abul Basar Lecturer, Department of Computer Science & Engineering(CSE) National Institute of Textile Engineering & Research(NITER) Mail : basar.cse@niter.edu.bd Mobile : +880 1758 443 472
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Personal Statement

I believe myself, to be the clear and logical mind with a practical approach to problem-solving and in cordiality, interested in knowledge through completion of all commitments. I wish to involve myself for doing such innovative work that will contribute to put my knowledge in practical use. I will also be dedicated by applying my skills to contribute towards the institution goal.

DECLARATION

I hereby declare that the details & information given above are complete and true to the best of my knowledge.



Signature